

MP352 – Special Relativity

Problem Set 3

Due by the end of lecture on Friday, 27 March 2015

1. Consider the coordinate transformation

$$\begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix} \mapsto \begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \Lambda \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$

for each of the following matrices:

$$\Lambda_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1/\sqrt{3} & \sqrt{2/3} \\ 0 & 0 & -\sqrt{2/3} & 1/\sqrt{3} \end{pmatrix}, \quad \Lambda_2 = \begin{pmatrix} 3/\sqrt{5} & 0 & 2/\sqrt{5} & 0 \\ 0 & 1 & 0 & 0 \\ 2/\sqrt{5} & 0 & 3/\sqrt{5} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

$$\Lambda_3 = \begin{pmatrix} \sqrt{2} & -1 & 0 & 0 \\ -1/\sqrt{2} & 1 & 0 & -1/\sqrt{2} \\ 0 & 0 & 1 & 0 \\ -1/\sqrt{2} & 1 & 0 & 1/\sqrt{2} \end{pmatrix}, \quad \Lambda_4 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Determine – using whatever methods you like – which of these transformations are Lorentz transformations.

2. Recall that all matrices in the Lorentz group must satisfy $\Lambda^T \eta \Lambda = \eta$, where

$$\eta = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

- (a) Show that this implies $\Lambda^{-1} = \eta^{-1} \Lambda^T \eta$.
- (b) Use the result from (a) to compute the inverse of each matrix in Problem 1 that gives a Lorentz transformations.
3. State whether each of the following pairs of sets and binary operations form a group. If “yes”, show that they satisfy the four basic properties which define a group. If “no”, state at least one of the properties that they do not have.

- (a) The set $\{1, -1/2 + i\sqrt{3}/2, -1/2 - i\sqrt{3}/2\}$ **and** multiplication.

- (b) The space of three-dimensional real vectors $\mathbb{R}^3$ **and** the dot product $\cdot$ between two vectors.
 - (c) The space of three-dimensional real vectors $\mathbb{R}^3$ **and** the cross product $\times$ between two vectors.
4. $GL(n, \mathbb{F})$ is the set of $n \times n$ matrices whose entries are in the number set $\mathbb{F}$ and which have nonzero determinant. $SL(n, \mathbb{F})$ are the matrices in $GL(n, \mathbb{F})$ which have determinant 1. If we take matrix multiplication to be the binary operation on both sets, explain why $GL(2, \mathbb{Z})$ is *not* a group but $SL(2, \mathbb{Z})$ *is* a group.